



# Incidence of SARS-COV-2 infection before and during pregnancy

A prospective cohort study based on electronic health records of pregnant women in England between 01 Jan 2020 to 18 Jun 2021

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## Introduction

- Pandemic caused by SARS-CoV-2 infected people of all age and gender, including pregnant women.
- As part of CVD-COVID-UK / COVID-IMPACT consortium, we established a cohort of pregnant women between 01 Jan 2020 and 18 Jun 2021 based on the electronic health records in England, to investigate the impact of pre-pregnancy infection on adverse outcomes.
- We aimed to estimate the incidence of first infection before and during pregnancy, and to evaluate how rates varied across subgroups.

## Methodology

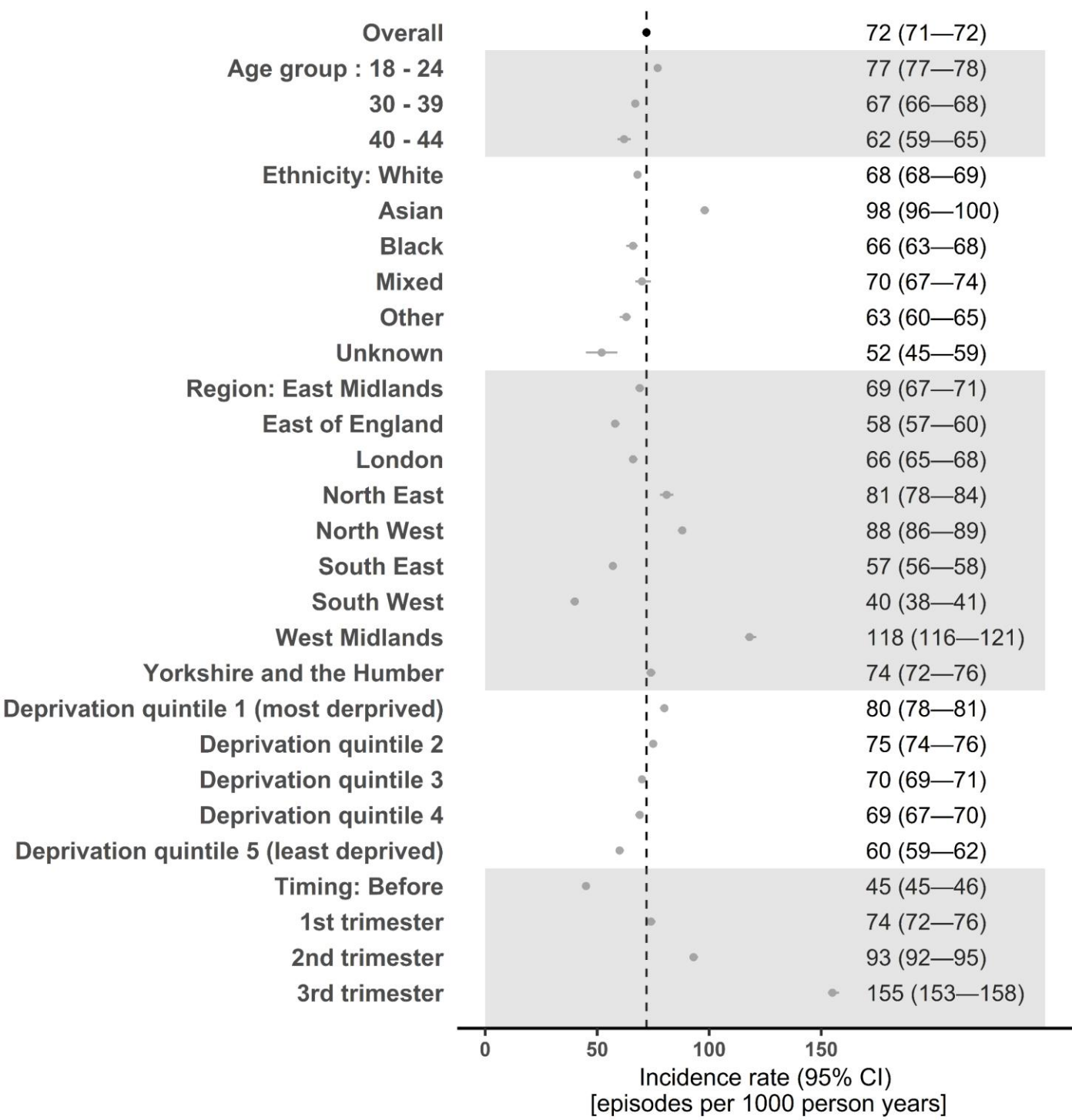
- Pregnant women aged 18–44 years
- Excluded if prior infection before 1 Jan 2020.
- Incidence rates (cases per 1,000 person-years) with 95% confidence intervals.
- Subgroup analyses by age, ethnicity, geographical region, deprivation level, and period of pregnancy

$$\text{Rate, } \lambda = \frac{\text{number of events}}{\text{total person-years of observation}} = \frac{d}{T}$$

## Results

- 672291 pregnancies were included in the study with 59240 (8.8%) infections at a rate of 72 infections per 1000 person-years (95% CI: 71–72).
- Trimester: Incidence was highest in the third trimester (155/1,000 PY) and lowest in the first trimester (74/1,000 PY).
- Age: 18–29 year age group showed the highest incidence (77/1,000 PY)
- Ethnicity: Rates were highest among those identifying as Asian or Asian British (98/1,000 PY)
- Region: West Midlands recorded the highest incidence (118/1,000 PY), while the South West had the lowest (40/1,000 PY)
- Deprivation: most deprived 80/1000 PY while least deprived 60/1000 PY.

Figure 1. Incidence rate per 1000 person years



## Conclusion

- Rate of SARS-CoV-2 infection before and during pregnancy in England was 72 episodes per 1000 person years between 01 Jan 2020 and 18 Jun 2021.
- Younger women, and women living in most deprived region experienced higher rate of infection
- Data suggest a higher rate of infection during the last trimester possibly coinciding with the peak of infection in the country

## References

Luisa et al, 2024, The lancet Regional Health, <https://doi.org/10.1016/j.lanepe.2024.101037>  
Sandoval et al, 2025, BMC medicine, <https://doi.org/10.1186/s12916-025-04094-y>  
Kirkwood & Sterne, 2003, Essential Medical Statistics, 2nd edition, Blackwell Science

